

SQL Basic Commands

Pizza Palace 🍕

Sample Tables Setup

Copy this whole dataset.

```
CREATE DATABASE pizza;
```

```
USE pizza;
```

```
CREATE TABLE pizzas (  
  pizza_id INT PRIMARY KEY,  
  pizza_name VARCHAR(50),  
  size ENUM('Small', 'Medium', 'Large', 'Extra Large'),  
  crust_type VARCHAR(30),  
  price DECIMAL(6,2),  
  calories INT,  
  category VARCHAR(20),  
  is_vegetarian BOOLEAN  
);
```

```
INSERT INTO pizzas VALUES
```

```
(1, 'Margherita', 'Medium', 'Thin Crust', 12.99, 280, 'Classic', TRUE),  
(2, 'Pepperoni Supreme', 'Large', 'Hand Tossed', 18.99, 350, 'Meat', FALSE),  
(3, 'Hawaiian Paradise', 'Medium', 'Stuffed Crust', 15.99, 320, 'Specialty', FALSE),  
(4, 'Veggie Delight', 'Small', 'Thin Crust', 10.99, 220, 'Vegetarian', TRUE),  
(5, 'Meat Lovers', 'Extra Large', 'Thick Crust', 22.99, 450, 'Meat', FALSE),  
(6, 'BBQ Chicken', 'Large', 'Hand Tossed', 19.99, 380, 'Specialty', FALSE),  
(7, 'Mushroom Magic', 'Medium', 'Thin Crust', 13.99, 260, 'Vegetarian', TRUE),  
(8, 'Spicy Italian', 'Large', 'Stuffed Crust', 20.99, 400, 'Specialty', FALSE),  
(9, 'Four Cheese', 'Small', 'Hand Tossed', 11.99, 310, 'Classic', TRUE),  
(10, 'Buffalo Chicken', 'Medium', 'Thick Crust', 16.99, 340, 'Specialty', FALSE),  
(11, 'Garden Fresh', 'Large', 'Thin Crust', 17.99, 290, 'Vegetarian', TRUE),  
(12, 'Supreme Deluxe', 'Extra Large', 'Hand Tossed', 24.99, 420, 'Specialty', FALSE);
```

```
CREATE TABLE customers (  
    customer_id INT PRIMARY KEY,  
    first_name VARCHAR(30),  
    last_name VARCHAR(30),  
    phone VARCHAR(15),  
    email VARCHAR(60),  
    city VARCHAR(30),  
    join_date DATE,  
    loyalty_points INT,  
    status ENUM('Regular', 'VIP', 'Gold', 'Inactive')  
);
```

```
INSERT INTO customers VALUES  
(201, 'Tony', 'Soprano', '555-0101', 'tony.soprano@email.com', 'Newark', '2023-01-15', 450,  
'VIP'),  
(202, 'Maria', 'Rossi', '555-0102', 'maria.rossi@email.com', 'Brooklyn', '2023-02-20', 280,  
'Regular'),  
(203, 'Giuseppe', 'Bianchi', '555-0103', 'giuseppe.b@email.com', 'Manhattan', '2022-11-10', 750,  
'Gold'),  
(204, 'Isabella', 'Ferrari', '555-0104', 'isabella.f@email.com', 'Queens', '2023-03-05', 120,  
'Regular'),  
(205, 'Marco', 'Lombardi', '555-0105', 'marco.lombardi@email.com', 'Bronx', '2022-08-18', 890,  
'Gold'),  
(206, 'Sofia', 'Romano', '555-0106', 'sofia.romano@email.com', 'Brooklyn', '2023-04-12', 65,  
'Regular'),  
(207, 'Antonio', 'Conti', '555-0107', 'antonio.conti@email.com', 'Staten Island', '2022-12-03', 340,  
'VIP'),  
(208, 'Francesca', 'Moretti', '555-0108', 'francesca.m@email.com', 'Manhattan', '2023-01-28',  
180, 'Regular'),  
(209, 'Luca', 'Esposito', '555-0109', 'luca.esposito@email.com', 'Queens', '2022-09-15', 520,  
'VIP'),  
(210, 'Giulia', 'Ricci', '555-0110', 'giulia.ricci@email.com', 'Bronx', '2023-05-22', 15, 'Inactive');
```

```
CREATE TABLE orders (  
    order_id INT PRIMARY KEY,  
    customer_id INT,  
    order_date DATE,  
    delivery_time TIME,  
    total_amount DECIMAL(8,2),  
    delivery_type ENUM('Pickup', 'Delivery', 'Dine-in'),  
    order_status ENUM('Pending', 'Preparing', 'Ready', 'Delivered', 'Cancelled'),  
    special_instructions TEXT  
);
```

INSERT INTO orders VALUES

(301, 201, '2024-01-15', '18:30:00', 45.97, 'Delivery', 'Delivered', 'Extra cheese on all pizzas'),
(302, 202, '2024-01-16', '12:15:00', 23.98, 'Pickup', 'Ready', 'No onions please'),
(303, 203, '2024-01-16', '19:45:00', 67.95, 'Dine-in', 'Delivered', NULL),
(304, 204, '2024-01-17', '20:00:00', 32.99, 'Delivery', 'Preparing', 'Call before delivery'),
(305, 205, '2024-01-17', '13:30:00', 18.99, 'Pickup', 'Ready', 'Extra spicy'),
(306, 206, '2024-01-18', '17:15:00', 28.98, 'Delivery', 'Delivered', NULL),
(307, 207, '2024-01-18', '19:00:00', 41.97, 'Dine-in', 'Cancelled', 'Table for 4'),
(308, 208, '2024-01-19', '14:45:00', 15.99, 'Pickup', 'Delivered', 'Light sauce'),
(309, 209, '2024-01-19', '21:30:00', 52.96, 'Delivery', 'Delivered', 'Ring doorbell twice'),
(310, 210, '2024-01-20', '16:20:00', 12.99, 'Pickup', 'Pending', NULL);

Assessment Questions

Section A: Basic SELECT Statements

- Q1.** Write a query to display all information about pizzas.
- Q2.** Display only the pizza_name, size, and price from the pizzas table.
- Q3.** Show all customer information from the customers table.
- Q4.** Display only the first_name, last_name, and city of all customers.
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Section B: WHERE Clause

- Q5.** Find all pizzas that are 'Large' size.
- Q6.** Display all customers from 'Brooklyn'.
- Q7.** Show all orders with total_amount greater than \$40.
- Q8.** Find all vegetarian pizzas (is_vegetarian = TRUE).
- Q9.** Display all customers with 'VIP' status.

Q10. Show all orders placed on '2024-01-17'.

Section C: LIKE Operator

Q11. Find all pizzas whose name starts with 'M'.

Q12. Display all customers whose first name ends with 'a'.

Q13. Show all pizzas that contain the word 'Chicken' in their name.

Q14. Find all customers whose email contains 'gmail' (Note: modify some emails if needed for practice).

Q15. Display all pizzas whose name contains 'Supreme'.

Section D: Comparison Operators

Q16. Find all pizzas with price between \$15 and \$20.

Q17. Display customers with loyalty points greater than or equal to 500.

Q18. Show all pizzas with calories less than 300.

Q19. Find orders with total_amount less than or equal to \$25.

Q20. Display all customers who joined before '2023-01-01'.

Section E: ORDER BY

Q21. Display all pizzas ordered by price in descending order.

Q22. Show all customers ordered by their loyalty points (highest first).

Q23. Display pizzas ordered by calories in ascending order.

Q24. Show customers ordered by city name alphabetically, then by last name.

Q25. Display orders ordered by order_date (newest first), then by total_amount (highest first).

Section F: LIMIT and OFFSET

Q26. Show the top 3 most expensive pizzas.

Q27. Display the 5 customers with the most loyalty points.

Q28. Show pizzas ranked 4-6 when ordered by price (use OFFSET).

Q29. Display the 3 most recent orders.

Q30. Show customers 3-5 when ordered by join date.

Section G: Aliases

Q31. Display pizza information with aliases: pizza_name as 'Pizza', size as 'Size', price as 'Cost'.

Q32. Show customer names with aliases: first_name as 'First Name', last_name as 'Last Name', loyalty_points as 'Points'.

Q33. Create a query showing customer's full name (concatenated) as 'Full Name' and their status as 'Membership Level'.

Q34. Display order information with aliases: order_id as 'Order #', total_amount as 'Total (\$)', delivery_type as 'Service Type'.

Section H: Combined Operations

Q35. Find the top 3 most expensive vegetarian pizzas.

Q36. Display the first 5 VIP customers ordered by loyalty points (highest first).

Q37. Show Large pizzas under \$20, ordered by calories, limit to 3 results.

Q38. Find customers from Brooklyn or Queens with more than 200 loyalty points, ordered by points.

Q39. Display the 2nd and 3rd cheapest pizzas in the 'Specialty' category.

Q40. Show orders from January 17-19, 2024, with delivery type 'Delivery', ordered by total amount.